

17_System of Linear Equations

A **system of linear equations** is a collection of **two or more linear equations** that involve the **same set of variables**.

Key Concepts

1. Linear Equation : A **linear equation** is an equation in which the highest power of each variable is **1**.

Example:

$$2x + 3y = 6$$

2. System : A **system** is formed when multiple linear equations are grouped together and solved simultaneously.

Example of a system with two variables:

$$\begin{cases} 2x + 3y = 6 \\ x - y = 4 \end{cases}$$

The solution is the set of values for the variables that satisfies **all equations at the same time**.

Type of solutions:

1. **Substitution Method**
2. **Cramer's Rule**
3. **Gaussian Elimination**

Applications of Systems of Linear Equations in Machine Learning

➤ **Linear Regression**

Solves the matrix equation

- $Xw = y$
- to find the **best-fit line or hyperplane** that minimizes prediction error.

➤ **Neural Networks**

Each layer performs a linear transformation of the form:

- $y = Wx + b$
- followed by a non-linear activation function.

➤ **Principal Component Analysis (PCA)**

Solves **homogeneous systems** to find **eigenvectors** of the covariance matrix, enabling **dimensionality reduction**.

➤ **Feature Selection**

Detects **linear dependence** among input variables to remove redundant features.

➤ **Support Vector Machines (SVM)**

Finds **optimal decision boundaries** using **linear constraints** and optimization techniques.

➤ **Gradient Descent**

Uses derivatives derived from linear systems to **optimize model parameters** efficiently.

➤ **Matrix Rank**

Indicates the **number of linearly independent equations or features** in a system.

➤ **Eigenvalues and Eigenvectors**

Found by solving the characteristic equation:

$$(A - \lambda I)v = 0$$

where λ represents eigenvalues and v represents eigenvectors.

What Is a Homogeneous System?

A **homogeneous system of linear equations** is a system in which **all constant terms are zero**.

This means **every equation is equal to 0**.

General Form

A homogeneous system with three variables can be written as:

$$\begin{cases} a_1x + b_1y + c_1z = 0 \\ a_2x + b_2y + c_2z = 0 \\ a_3x + b_3y + c_3z = 0 \end{cases}$$

Example

$$\begin{cases} x + y + z = 0 \\ 2x - y + z = 0 \\ x + 2y - 3z = 0 \end{cases}$$

How to Solve Systems of Linear Equations

1. Substitution Method (Two Variables)

Given the system:

$$\begin{cases} x + y = 7(1) \\ 3x - y = 5(2) \end{cases}$$

Step 1: Solve Equation (1) for one variable

$$x = 7 - y$$

Step 2: Substitute into Equation (2)

$$\begin{aligned} 3(7 - y) - y &= 5 \\ \Rightarrow 21 - 3y - y &= 5 \\ \Rightarrow 21 - 4y &= 5 \\ \Rightarrow -4y &= 5 - 21 = -16 \\ \Rightarrow y &= 4 \end{aligned}$$

Step 3: Find x

$$x = 7 - y = 7 - 4 = 3$$

 **Final Answer:**

$$x = 3, y = 4$$

Given System of Equations

$$\begin{cases} x + 2y - z = 4 \dots\dots\dots (1) \\ 2x - y + 3z = 1 \dots\dots\dots (2) \\ -x + 3y + 2z = 7 \dots\dots\dots (3) \end{cases}$$

Goal

Find the values of x , y , **and** z that satisfy **all three equations simultaneously**.

Step 1: Solve Equation (1) for x

From (1):

$$x = 4 - 2y + z$$

Step 2: Substitute x into Equations (2) and (3)

Substitute into Equation (2)

$$\begin{aligned} 2(4 - 2y + z) - y + 3z &= 1 \\ \Rightarrow 8 - 4y + 2z - y + 3z &= 1 \\ \Rightarrow 8 - 5y + 5z &= 1 \\ \Rightarrow -5y + 5z &= -7 \\ \Rightarrow -y + z &= -\frac{7}{5} \dots\dots\dots (4) \end{aligned}$$

Substitute into Equation (3)

$$\begin{aligned} -(4 - 2y + z) + 3y + 2z &= 7 \\ \Rightarrow -4 + 2y - z + 3y + 2z &= 7 \\ \Rightarrow -4 + 5y + z &= 7 \\ \Rightarrow 5y + z &= 11 \dots\dots\dots (5) \end{aligned}$$

Step 3: Solve the two-variable system

From (4):

$$z = y - \frac{7}{5}$$

Substitute into (5):

$$\begin{aligned} 5y + \left(y - \frac{7}{5}\right) &= 11 \\ \Rightarrow 6y - \frac{7}{5} &= 11 \\ \Rightarrow 6y &= \frac{62}{5} \\ \Rightarrow y &= \frac{31}{15} \end{aligned}$$

Step 4: Find z

$$\begin{aligned} z &= y - \frac{7}{5} \\ \Rightarrow z &= \frac{31}{15} - \frac{21}{15} = \frac{10}{15} = \frac{2}{3} \end{aligned}$$

Step 5: Find x

$$\begin{aligned} x &= 4 - 2y + z \\ \Rightarrow x &= 4 - 2\left(\frac{31}{15}\right) + \frac{2}{3} \\ \Rightarrow x &= \frac{60 - 62 + 10}{15} = \frac{8}{15} \end{aligned}$$

✅ **Final Answer**

$$\boxed{x = \frac{8}{15}, y = \frac{31}{15}, z = \frac{2}{3}}$$